Taiwan Cancer Registry

Sex-Specific Cancer Atlas — 78,441 Patients · 37 Sites · 2003–2020

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Overview

Sex is the strongest binary covariate in the Taiwan Cancer Registry.
The VAE identified two orthogonal sex-linked axes:
  z4 – UADT/field-cancerization (rank-biserial=−0.621, male)
  z0/z5 – Hormonal/gynecologic  (rbi=+0.163/+0.409, female)

These axes map to distinct carcinogenic exposures:
  Male   → betel nut + tobacco (UADT) + HBV/metabolic syndrome (GI)
  Female → endogenous hormones (breast, endometrial, ovarian)

Key Findings (all FDR<0.05)

  Site sex-OR spectrum:
    Male-dominant:   23 sites
    Female-dominant: 1 sites
    Sex-neutral:     13 sites

  Strongest male site:   C12 pyriform  OR=40×
  Strongest female site: C50 breast    OR=0.002×

  Age × sex interaction:
    UADT males present 5–9yr YOUNGER than UADT females
    (C06: Δ=−9yr, C02: Δ=−7yr, C15: Δ=−5yr)
    C50 breast: males 15yr OLDER than females (incidental/late-onset)

  Multi-cancer rate:
    Males: 6.8%  vs  Females: 3.4%  (2.0× difference)
    UADT males: 14.9% vs UADT females: 7.7%  (axis-specific)

  Survival:
    Log-rank M vs F: p<0.001; males worse overall
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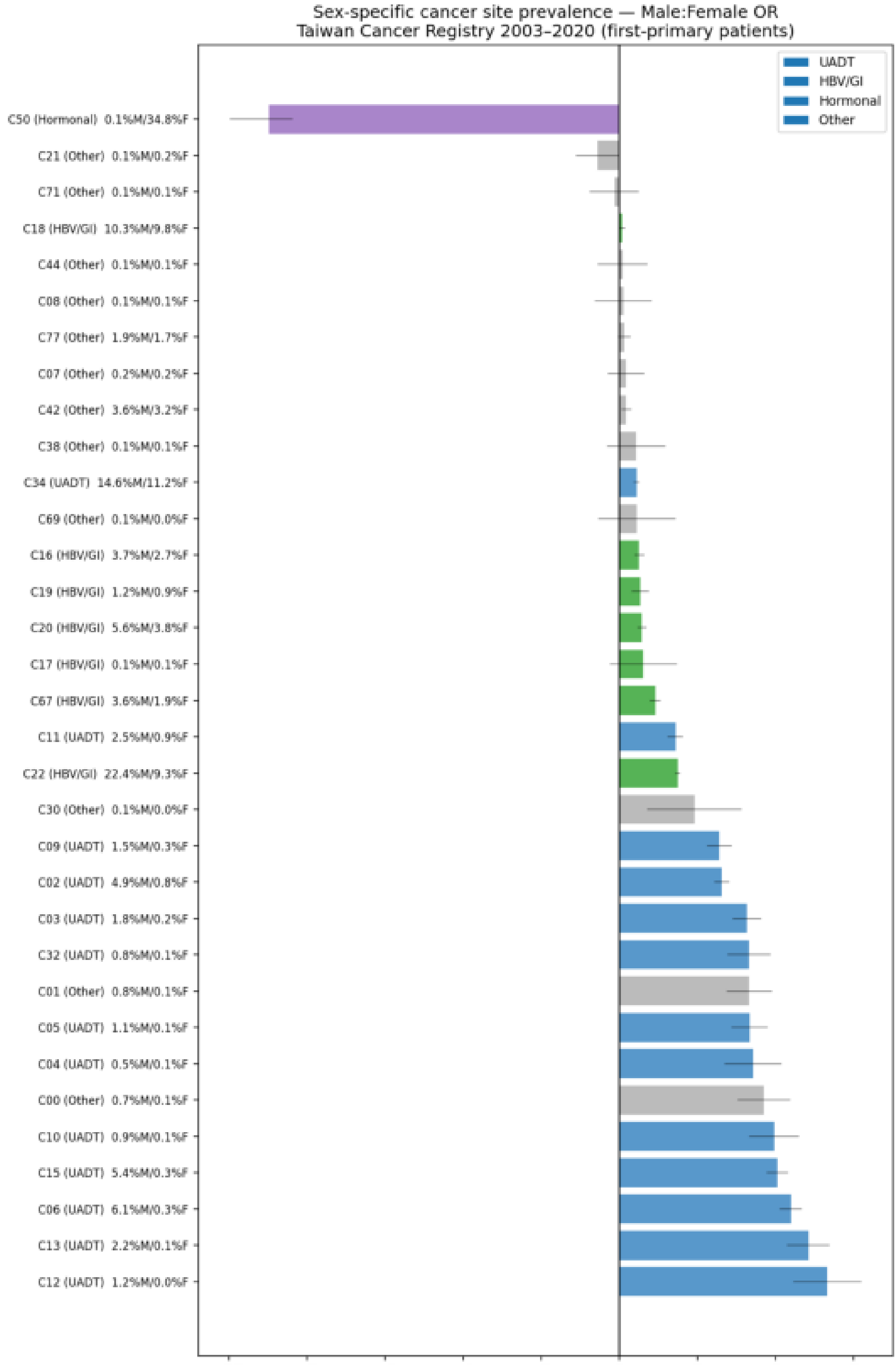
Sex-specific cancer prevalence — Male:Female OR forest

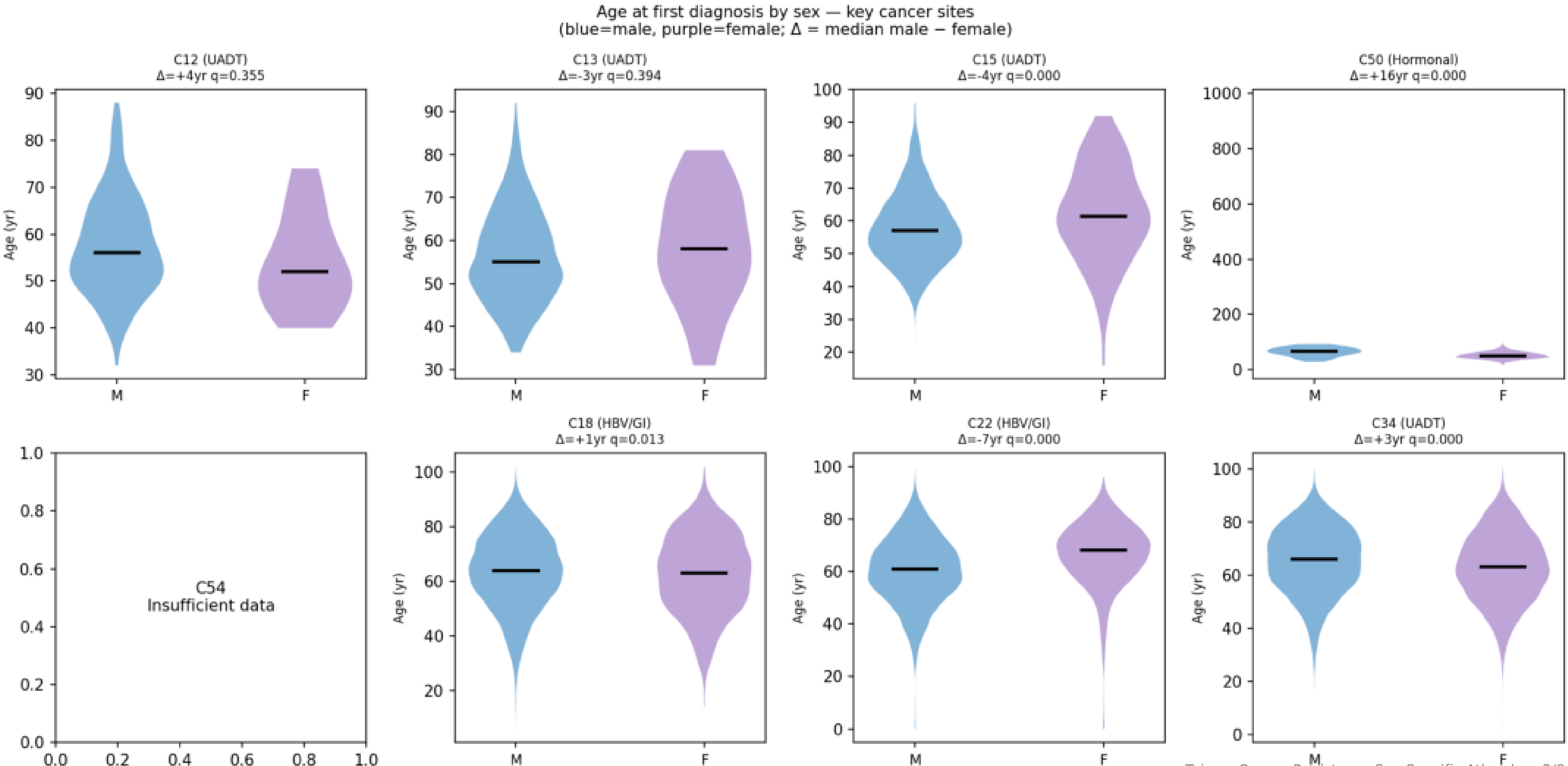
Male:Female OR by site — log₂ scale (all 37 sites)



log₂(Male:Female odds ratio) ← female dominant | male dominant →

Age at first diagnosis by sex — UADT males present younger

Age at first diagnosis by sex — key sites (blue=male, purple=female)



Age at first diagnosis: sex differences by site

Age × sex differences by site (FDR<0.05, top 12)

Site	Axis	M median	F median	Δ (M−F)	q FDR
C69	Other	64	46	+18	0.0344
C50	Hormonal	66	51	+16	0.0000
C00	Other	56	69	-13	0.0037
C03	UADT	56	66	-10	0.0001
C08	Other	61	51	+10	0.0117
C06	UADT	52	61	-9	0.0000
C02	UADT	51	58	-7	0.0000
C22	HBV/GI	61	68	-7	0.0000
C15	UADT	57	62	-4	0.0004
C34	UADT	66	63	+3	0.0000
C16	HBV/GI	66	63	+3	0.0004
C67	HBV/GI	67	70	-2	0.0096

Age × sex — biological interpretation

UADT sites: males present YOUNGER

C06 oral cavity: −9yr C02 tongue: −7yr
C15 esophagus: −5yr C03 floor of mouth: −10yr

Interpretation: heavy betel nut + tobacco exposure begins in adolescence/early adulthood in Taiwan males. Earlier cumulative exposure → earlier cancer onset. Female UADT patients likely have lower betel/tobacco exposure → present later.

C22 liver HCC: males −7yr younger

Consistent with higher HBV carrier rate in males (male HBV clearance is lower → higher chronicity).

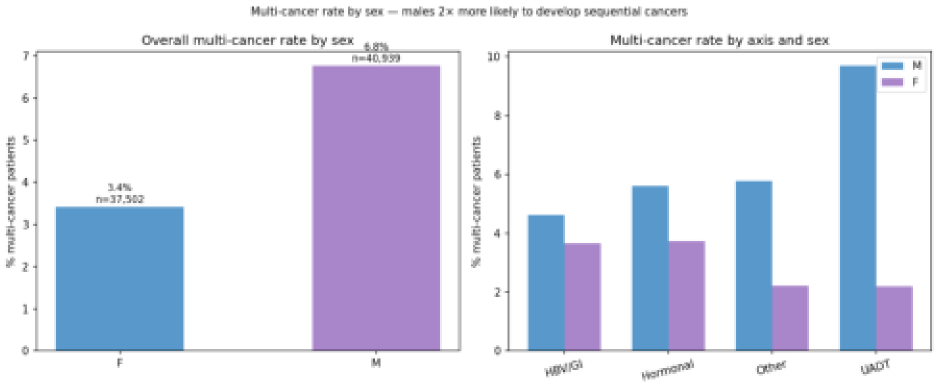
C50 breast: males +15yr older

Male breast cancer is rare, late-onset, typically post-60yr in Taiwanese men. Female breast cancer peaks 40–55yr (younger than Western populations).

Clinical implication:
Screening programmes should be sex-stratified and use different age-at-initiation thresholds.

Males 2× more likely to develop sequential cancers

Multi-cancer rate by sex (overall + by axis)



Multi-cancer rate by sex and axis

Multi-cancer rate by sex — key finding

Overall:
Males: 6.8%   Females: 3.4%   Ratio: 2.0×

By carcinogenic axis:
UADT M: 14.9% vs F: 7.7%   Ratio: 1.9×
HBV/GI M: 4.2%   vs F: 3.0%   Ratio: 1.4×
Other   M: 6.0%   vs F: 2.1%   Ratio: 2.9×

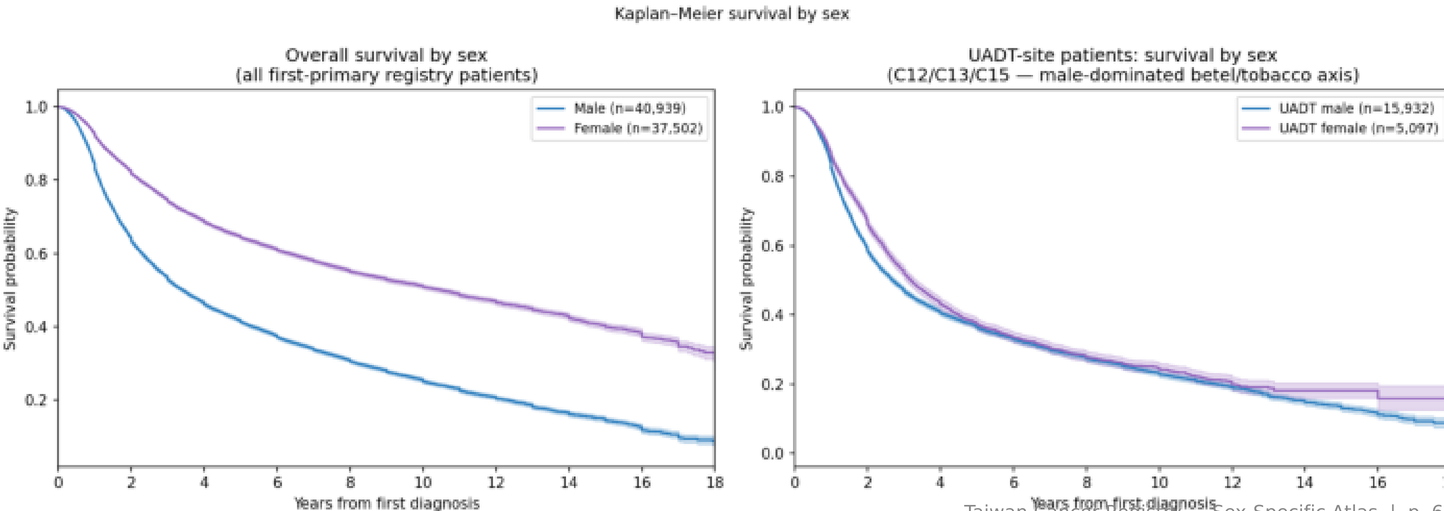
Interpretation:
The 2× male excess in multi-cancer rate is not simply explained by UADT field cancerization (the largest contributor) — Other-axis males also show 2.9× excess, suggesting a systemic sex difference in multi-primary risk.

Possible mechanisms:
1. Betel nut + alcohol + tobacco act synergistically across multiple anatomical sites (not only UADT)
2. Immune surveillance differences between sexes: higher female cytotoxic T-cell activity may suppress nascent second primaries more effectively
3. Hormonal influence on DNA repair capacity

Note: hormonal-axis males n=40 only (prostate + lung) — UADT dominates male carcinogenesis.

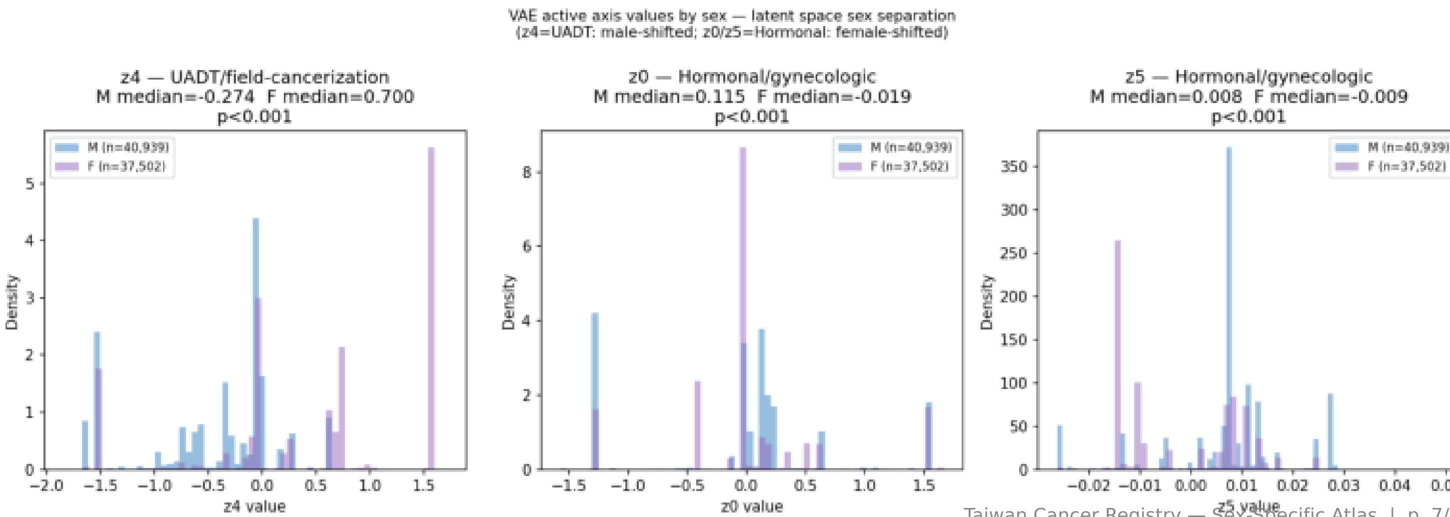
Survival by sex — overall and UADT-specific

Kaplan–Meier survival by sex (overall + UADT subset)



VAE latent axes encode sex — z4 male-shifted, z0/z5 female-shifted

VAE active axis values by sex — latent space sex separation



Conclusions and Limitations

Sex-Specific Cancer Atlas — Taiwan Cancer Registry 2003–2020

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Principal Conclusions

1.  The Taiwan cancer registry is strongly sex-stratified.
    Of 37 sites, 23 are male-dominant, 1 are female-dominant,
    and 13 are sex-neutral — all FDR<0.05.

2.  Two orthogonal sex-linked carcinogenic axes:
    MALE   → betel/tobacco UADT (z4, rbi=−0.621) + HBV/GI
    FEMALE → hormonal/gynecologic (z0/z5, rbi=+0.409)
    These axes explain the bulk of sex heterogeneity in the registry.

3.  UADT males present 5–9yr younger than females for the same site.
    Consistent with earlier-onset betel nut exposure in male adolescents.
    Screening programmes should use sex-specific age-at-initiation.

4.  Males have 2× the multi-cancer rate of females (6.8% vs 3.4%).
    Excess is not confined to UADT — 'Other' sites show 2.9× excess.
    Possible mechanisms: carcinogen synergy, immune sex differences.

5.  Survival is significantly worse in males (log-rank p<0.001).
    Partly explained by UADT predominance (high-mortality sites)
    and later-stage presentation.

Limitations

   No betel nut/tobacco/alcohol exposure data — sex differences are
     attributed to exposure axes by ecological inference only.
   No hormonal data (parity, OCP, HRT) for female hormonal axis.
   Registry records first-primary site per patient; rare non-primary
     sex assignment errors possible for sex-exclusive sites.
   Multi-cancer rate may be partially confounded by follow-up duration
     (males have higher baseline mortality → shorter FU window for
     detecting second cancers).

Next Steps

   Sex-stratified survival models for each axis (Cox with sex × axis interaction)
   Age-standardised incidence rates by site and sex (vs general population)
   Link sex-specific multi-cancer risk to Transformer surveillance calendar
     (male UADT patients warrant more aggressive 6-monthly endoscopy)
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